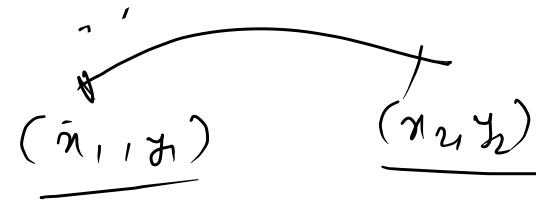


Rectification  
(To find length of curve)  $x = f(y)$

length of curve =  $S$

① If  $y = f(x)$

$$S = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} \underline{\underline{dx}}$$



② If  $x = f(y)$

$$S = \int_{y_1}^{y_2} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

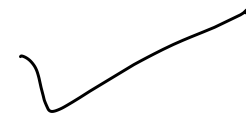
③ If  $x = f_1(t)$  &  $y = f_2(t) \rightarrow$  Parametric eq's

$$S = \int_{t_1}^{t_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \underline{\underline{dt}}$$

polar

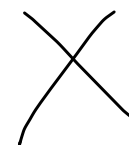
④ If  $r = f(\theta)$

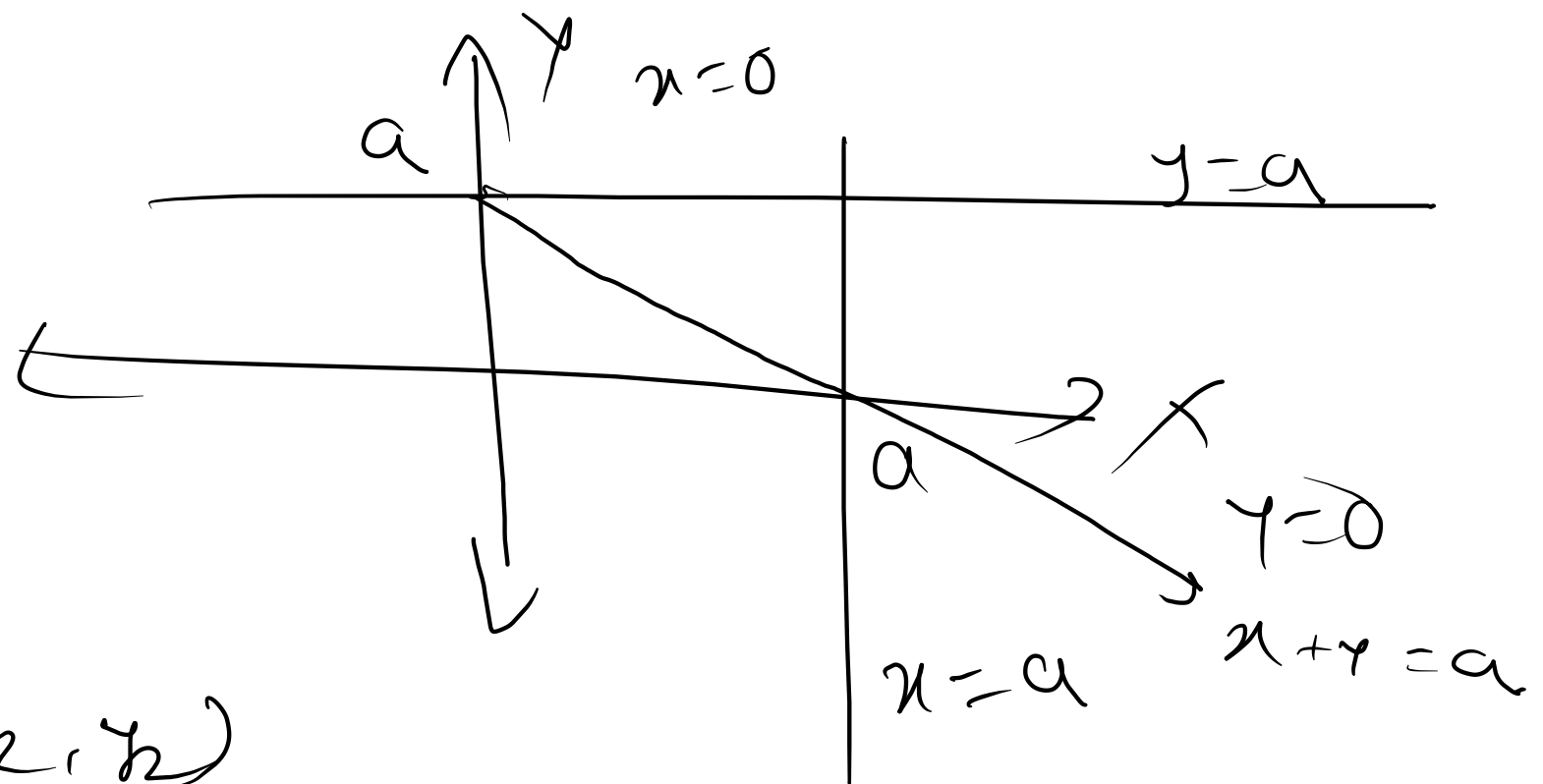
$$S = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$



⑤ If  $\theta = f(r)$

$$S = \int_{r_1}^{r_2} \sqrt{1 + r^2 \left(\frac{d\theta}{dr}\right)^2} dr$$





$(x_1, y_1)$   $(x_2, y_2)$

$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1} \quad \left\{ \checkmark \right.$$

$$x = \pm \sqrt{y}$$

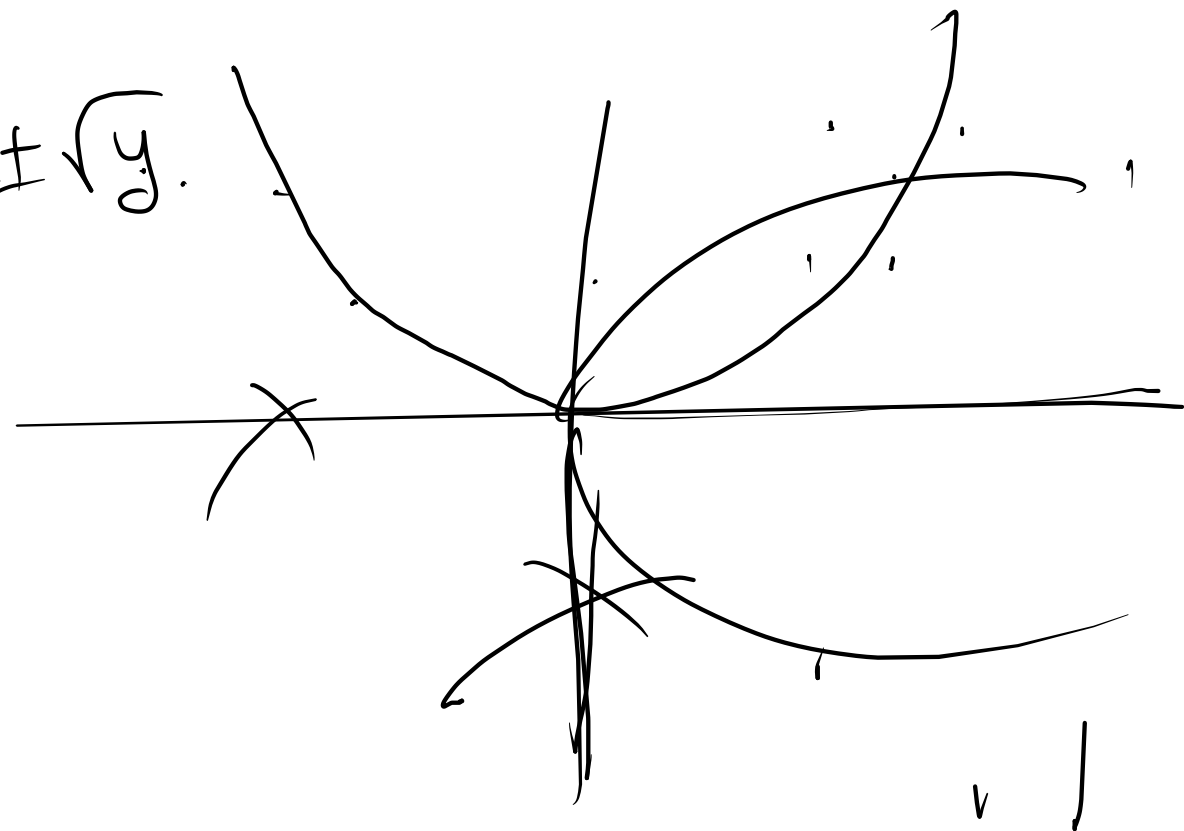
$$y = x^2$$

$$x = y^2$$

$$y = \pm \sqrt{x}$$

$$(x-a)^2 + (y-b)^2 = r^2$$

center  $(a, b)$   
rad.  $r$



Find the total length of the curve  $x^{2/3} + y^{2/3} = a^{2/3}$  Astroid  
4-cusp hypocycloid.

$S =$  Total length

$$= 4 \, l(AB)$$

$$= 4 \int_a^0 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{--- } y=f(x)$$

$$x^{2/3} + y^{2/3} = a^{2/3} \quad \text{--- } (1)$$

diff wrt.  $x$

$$\frac{2}{3} x^{-1/3} + \frac{2}{3} y^{-1/3} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x^{1/3}}{y^{1/3}} = -\frac{y^{2/3}}{x^{2/3}}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{y^{4/3}}{x^{4/3}}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{y^{4/3}}{x^{4/3}} = \frac{x^{4/3} + y^{4/3}}{x^{4/3}}$$

$$= \frac{a^{4/3}}{x^{4/3}} \quad \text{--- from (1)}$$

$$S = 4 \int_a^0 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

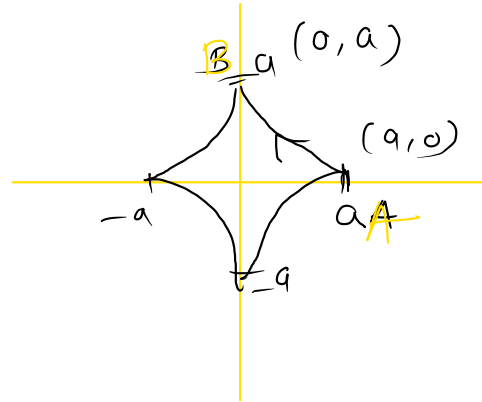
$$= 4 \int_a^0 \sqrt{\frac{a^{4/3}}{x^{4/3}}} dx = 4 a^{1/3} \int_a^0 x^{-1/3} dx$$

$$= 4 a^{1/3} \left[ \frac{x^{2/3}}{2/3} \right]_a^0$$

$$= 4 a^{1/3} \left[ 0 - a^{2/3} \cdot \frac{3}{2} \right]$$

$$= -\frac{4 \times 3}{2} a = -6a$$

Total length of  
 given curve = 6a



$$\int \sqrt{x^2 + a^2} dx = \frac{x}{2} \sqrt{x^2 + a^2} + \frac{a^2}{2} \log(x + \sqrt{x^2 + a^2})$$

Find the length of the parabola  $x^2 = 4y$  which lies inside the circle  $x^2 + y^2 = 6y$ .

$$\left. \begin{aligned} x^2 + y^2 &= 6y \end{aligned} \right\}$$

$$\underline{x^2 + y^2 - 6y = 0}$$

$$x^2 + y^2 - 6y + 9 = 9$$

$$x^2 + (y-3)^2 = 3^2$$

center  $(0, 3)$  rad 3

$$x^2 + y^2 = 6y \quad x^2 = 4y$$

put  $x^2 = 4y$

$$4y + y^2 = 6y \Rightarrow y^2 - 2y = 0 \quad y(y-2) = 0 \Rightarrow y = 0 \text{ or } y = 2$$

as  $x^2 = 4y$  for  $y = 0 \Rightarrow x = 0$

for  $y = 2 \Rightarrow x^2 = 8 \Rightarrow x = \pm 2\sqrt{2}$

$$(0, 0) \quad (2\sqrt{2}, 2) \quad (-2\sqrt{2}, 2)$$

$$S = \text{Total length} = l(AB)$$

$$= 2l(OB)$$

$$= 2 \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad \text{--- (y=f(x))}$$

$$x^2 = 4y$$

$$\Rightarrow y = \frac{x^2}{4} \Rightarrow \frac{dy}{dx} = \frac{2x}{4} = \frac{x}{2}$$

$$S = 2 \int_0^{2\sqrt{2}} \sqrt{1 + \left(\frac{x}{2}\right)^2} dx$$

$$= 2 \int_0^{2\sqrt{2}} \sqrt{\frac{4+x^2}{4}} dx$$

$$= \frac{2}{2} \int_0^{2\sqrt{2}} \sqrt{x^2 + 4} dx$$

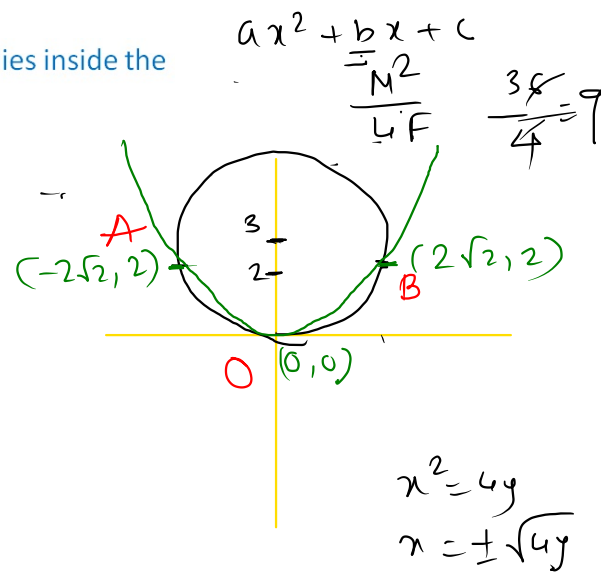
$$= \left[ \frac{x}{2} \sqrt{x^2 + 4} + \frac{4}{2} \log(x + \sqrt{x^2 + 4}) \right]_0^{2\sqrt{2}}$$

$$= \frac{2\sqrt{2}}{2} \sqrt{12} + 2 \log(2\sqrt{2} + \sqrt{12}) - 2 \log 2$$

$$= 2\sqrt{6} + 2 \log(2\sqrt{2} + 2\sqrt{3}) - 2 \log 2$$

$$= 2\sqrt{6} + 2 \log\left(\frac{2\sqrt{2} + 2\sqrt{3}}{2}\right)$$

$$= 2\sqrt{6} + 2 \log(\sqrt{2} + \sqrt{3})$$



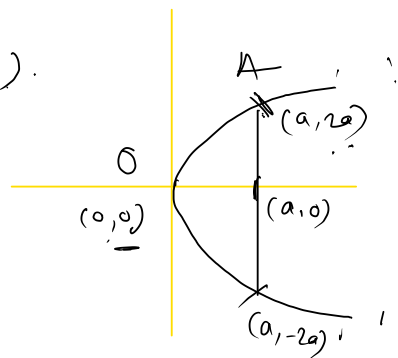


Hw Show that the length of the parabola  $y^2 = 4ax$  from the vertex to the end of the latus rectum is  $a[\sqrt{2} + \log(1 + \sqrt{2})]$ .  
Find the length of the arc cut off by the line  $3y = 8x$ .

$$y^2 = 4ax \Rightarrow x = \frac{y^2}{4a} = f(y)$$

$$y = \pm \sqrt{4ax}$$

$$x=a \quad y = \pm \sqrt{4a^2} = \pm 2a$$



$$S = l(OA)$$

$$= \int_{-2a}^{2a} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$\text{as } x = \frac{y^2}{4a} \quad \frac{dx}{dy} = \frac{2y}{4a} = \frac{y}{2a}$$

$$= \int_0^{2a} \sqrt{1 + \left(\frac{y}{2a}\right)^2} dy = \frac{1}{2a} \int_0^{2a} \sqrt{(2a)^2 + y^2} dy$$

$$= \frac{1}{2a} \left[ \frac{y}{2} \sqrt{(2a)^2 + y^2} + \frac{(2a)^2}{2} \log(y + \sqrt{(2a)^2 + y^2}) \right]_0^{2a}$$

$$= \frac{1}{2a} \left[ \frac{2a}{2} \sqrt{8a^2} + \frac{2a^2}{2} \log(2a + \sqrt{8a^2}) - \frac{4a^2}{2} \log(2a) \right]$$

$$= \frac{1}{2a} \left[ \underbrace{a^2 2\sqrt{2}} + \underbrace{2a^2} \log(2a + 2a\sqrt{2}) - \underbrace{2a^2} \log 2a \right]$$

$$= \frac{2a^2}{2a} \left[ \sqrt{2} + \log\left(\frac{2a + 2a\sqrt{2}}{2a}\right) \right] = a[\sqrt{2} + \log(1 + \sqrt{2})]$$

$$y^2 = 4ax$$

$$64x^2 = 4ax$$

$$64x^2 - 4ax = 0 \Rightarrow x(64x - 4a) = 0 \Rightarrow x=0, x = \frac{4a}{64} = \frac{a}{16}$$

$$y = \frac{8x}{3}$$

$$3y = 8x$$

$$y = \frac{8x}{3}$$

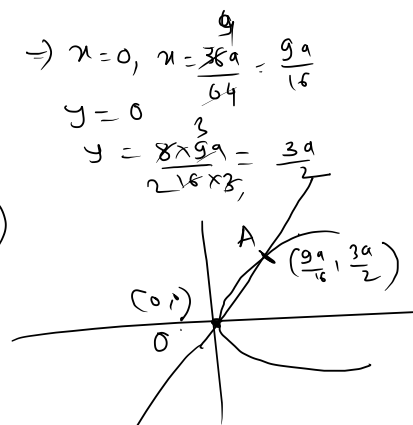
$$\therefore \text{for } x=0 \quad y = \frac{9a}{16}$$

$$(0,0) \quad \left(\frac{9a}{16}, \frac{3a}{2}\right)$$

$$S = l(OA)$$

$$= \int_0^{\frac{3a}{2}} \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$= \frac{1}{2a} \int_0^{\frac{3a}{2}} \sqrt{(2a)^2 + y^2} dy$$



H.W.

Ans  $\frac{15}{16}a + a \log 2$

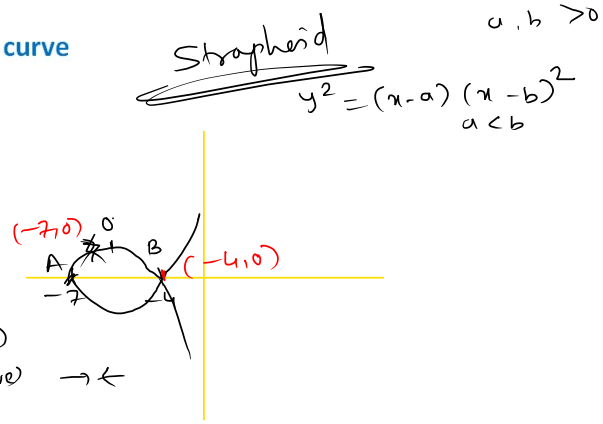
❖ Find the total length of the loop of the curve

$9y^2 = (x+7)(x+4)^2$

So  $y=0 \implies (x+7)(x+4)^2=0$

$(-7,0) \quad (-4,0)$   
 $x = -7, -4$

$x < -7$   
 $x+7 < 0$  (-ve)  
 $y^2 = -ve$  (+ve)  $\rightarrow \leftarrow$



$S = \text{length of loop} = 2 \times \text{length of upper half of loop}$   
 $= 2 \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$

So  $y^2 = (x+7)(x+4)^2$  ( $y=f(x)$ )  
diff wrt  $x$

$18y \frac{dy}{dx} = (x+7) 2(x+4) + (x+4)^2$   
 $= (x+4) [2x+14 + x+4] = (x+4)(3x+18)$

$18y \frac{dy}{dx} = 3(x+4)(x+6)$

$\frac{dy}{dx} = \frac{(x+4)(x+6)}{6y} \implies \left(\frac{dy}{dx}\right)^2 = \frac{(x+4)^2(x+6)^2}{36y^2}$   
 $= \frac{(x+4)^2(x+6)^2}{36(x+7)(x+4)^2}$   
 $= \frac{(x+6)^2}{4(x+7)}$

$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{(x+6)^2}{4(x+7)} = \frac{4(x+7) + (x+6)^2}{4(x+7)}$   
 $= \frac{4x+28 + x^2+12x+36}{4(x+7)}$   
 $= \frac{x^2+16x+64}{4(x+7)} = \frac{(x+8)^2}{4(x+7)}$

$S = 2 \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2 \int_{-7}^{-4} \sqrt{\frac{(x+8)^2}{4(x+7)}} dx$   
 $= \frac{2}{2} \int_{-7}^{-4} \frac{x+8}{\sqrt{x+7}} dx$

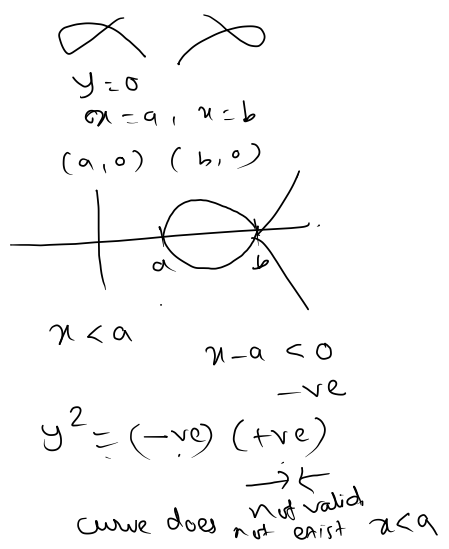
put  $x+7=t \implies x=t-7$   $\left| \begin{matrix} x: -7 \rightarrow -4 \\ t: 0 \rightarrow 3 \end{matrix} \right.$   
 $\frac{dx}{dt} = 1$

$= \int_0^3 \left( \frac{t-7+8}{t^{1/2}} \right) dt = \int_0^3 \frac{t+1}{t^{1/2}} dt$

$= \int_0^3 t^{1/2} + t^{-1/2} dt = \left[ \frac{t^{3/2}}{3/2} + \frac{t^{1/2}}{1/2} \right]_0^3$

$= \frac{2}{3} 3^{3/2} + 2\sqrt{3} = \frac{2}{3} (3\sqrt{3}) + 2\sqrt{3}$

$= \frac{2 \times 3\sqrt{3}}{3} + 2\sqrt{3} = 4\sqrt{3}$



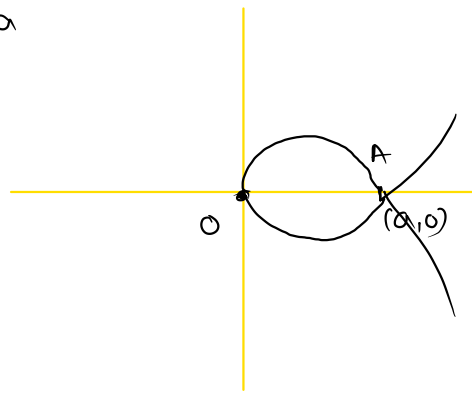
• HW Find the length of the loop  $3ay^2 = x(x-a)^2$

$$y = 0 \Rightarrow x = 0, x = a$$

$S$  = length of loop

= 2 length of upper half of loop

$$= 2 \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$



$$3ay^2 = x(x-a)^2$$

Diff w.r.t  $x$ .

$$6ay \frac{dy}{dx} = x \cdot 2(x-a) + (x-a)^2$$

$$= (x-a)(2x + x - a)$$

$$= (x-a)(3x-a)$$

$$\frac{dy}{dx} = \frac{(x-a)(3x-a)}{6ay} \Rightarrow \left(\frac{dy}{dx}\right)^2 = \frac{(x-a)^2(3x-a)^2}{36a^2y^2}$$

$$= \frac{(x-a)^2(3x-a)^2}{36a^2 \frac{x(x-a)^2}{3a}}$$

$$\left(\frac{dy}{dx}\right)^2 = \frac{(3x-a)^2}{12ax}$$

$$1 + \left(\frac{dy}{dx}\right)^2 = 1 + \frac{(3x-a)^2}{12ax} = \frac{12ax + 9x^2 - 6xa + a^2}{12ax}$$

$$= \frac{9x^2 + 6ax + a^2}{12ax} = \frac{(3x+a)^2}{12ax}$$

$$S = 2 \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = 2 \int_0^a \sqrt{\frac{(3x+a)^2}{12ax}} dx$$

$$= 2 \int_0^a \frac{(3x+a)}{\sqrt{12a} \sqrt{x}} dx$$

$$= \frac{2}{2\sqrt{3a}} \int_0^a \left( \frac{3x}{x^{1/2}} + \frac{a}{x^{1/2}} \right) dx$$

$$= \frac{1}{\sqrt{3a}} \int_0^a (3x^{1/2} + ax^{-1/2}) dx$$

$$= \frac{1}{\sqrt{3a}} \left[ \frac{3x^{3/2}}{3/2} + ax^{1/2} \right]_0^a$$

$$= \frac{1}{\sqrt{3a}} [2a^{3/2} + 2a^{3/2}] = \frac{1}{\sqrt{3a}} 4a^{3/2}$$

$$= \frac{4a^{3/2}}{\sqrt{3} a^{1/2}} = \frac{4a}{\sqrt{3}}$$

Prove that the length of the arc of the curve  $y = \log \left( \frac{e^x - 1}{e^x + 1} \right)$  from  $x = 1, x = 2$  is  $\log \left( e + \frac{1}{e} \right)$

$$y = \log \left( \frac{e^x - 1}{e^x + 1} \right)$$

$$= \log(e^x - 1) - \log(e^x + 1)$$

$$\frac{dy}{dx} = \frac{e^x}{e^x - 1} - \frac{e^x}{e^x + 1}$$

$$= \frac{e^{2x} + e^x - (e^{2x} - e^x)}{e^{2x} - 1} = \frac{2e^x}{e^{2x} - 1}$$

$$1 + \left( \frac{dy}{dx} \right)^2 = 1 + \frac{4e^{2x}}{(e^{2x} - 1)^2} = \frac{(e^{2x})^2 - 2e^{2x} + 1 + 4e^{2x}}{(e^{2x} - 1)^2}$$

$$= \frac{(e^{2x})^2 + 2e^{2x} + 1}{(e^{2x} - 1)^2} = \frac{(e^{2x} + 1)^2}{(e^{2x} - 1)^2}$$

$$S = \int_1^2 \frac{e^{2x} + 1}{e^{2x} - 1} dx$$

$$= \int_1^2 \frac{e^{-x}}{e^x} \frac{(e^{2x} + 1)}{(e^{2x} - 1)} dx$$

$$= \int_1^2 \frac{e^x + e^{-x}}{e^x - e^{-x}} dx = \left[ \log(e^x - e^{-x}) \right]_1^2 = \left[ \log(e^2 - e^{-2}) - \log(e - e^{-1}) \right]$$

$$(e^x - e^{-x}) = e^x - (-e^{-x})$$

$$= \log \left( \frac{e^2 - \frac{1}{e^2}}{e - \frac{1}{e}} \right)$$

$$= \log \left( \frac{(e + \frac{1}{e})(e - \frac{1}{e})}{(e - \frac{1}{e})} \right)$$

$$y = f(x)$$

$$S = \int_a^b \sqrt{1 + \left( \frac{dy}{dx} \right)^2} dx$$

$$= \int_1^2 \sqrt{1 + \left( \frac{dy}{dx} \right)^2} dx \quad \text{--- (1)}$$

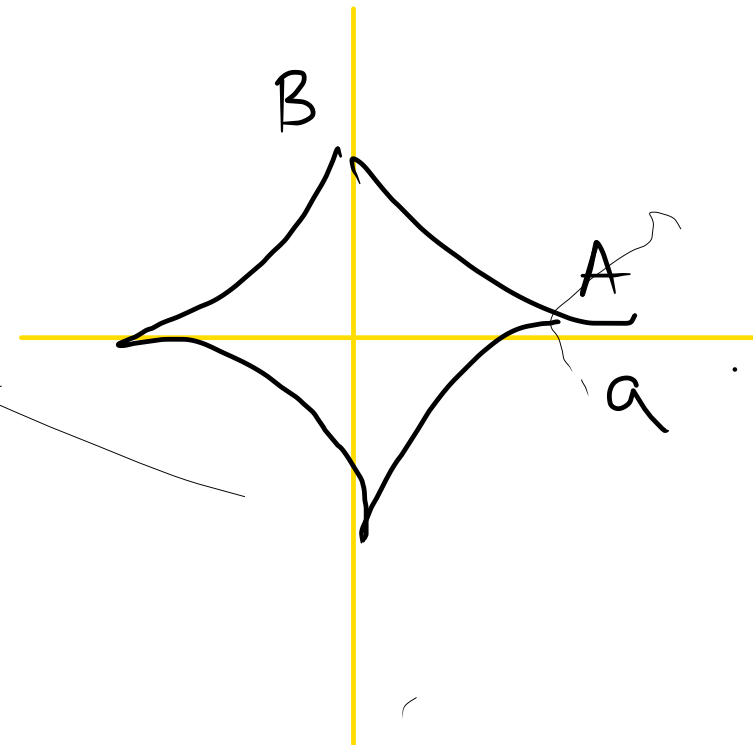
$$\int \frac{f'(x)}{f(x)} dx = \log f(x)$$

$$x^{2/3} + y^{2/3} = a^{2/3}$$

Find the length of the asteroid  $x = a \cos^3 t$ ,  $y = a \sin^3 t$ .

$$S = 4 \, l(AB)$$

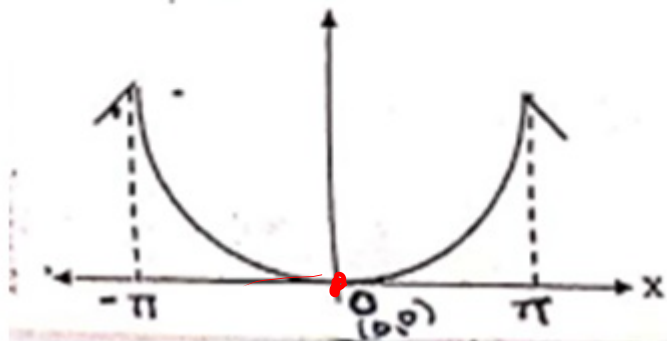
$$= 4 \int_0^{\pi/2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$



$6a$

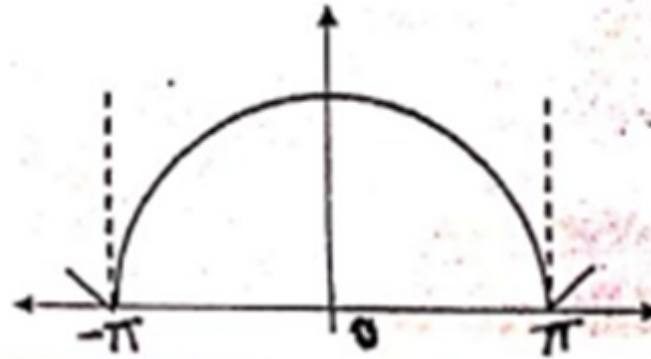
$$(1) \begin{cases} x = a(\theta + \sin \theta) \\ y = a(1 - \cos \theta) \end{cases}$$

$\theta$   $x$   $y$   
0 0 0



$$(2) \begin{cases} x = a(\theta + \sin \theta) \\ y = a(1 + \cos \theta) \end{cases}$$

$\theta$   $x$   $y$   
0 0 2



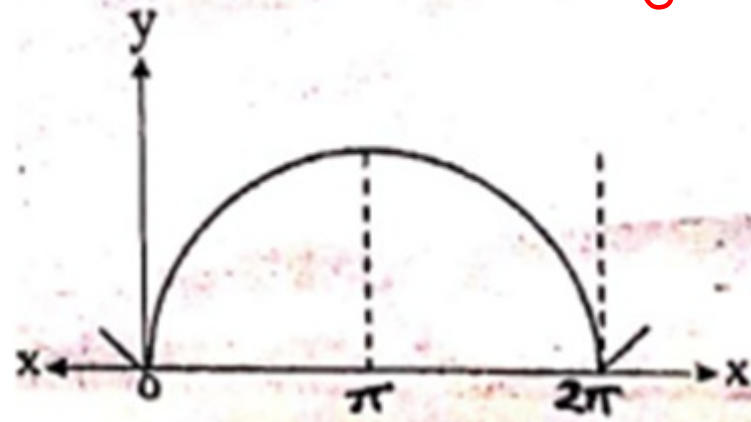
$$(1) \begin{cases} x = a(\theta - \sin \theta) \\ y = a(1 + \cos \theta) \end{cases}$$

$\theta$   $x$   $y$   
0 0 2



$$(3) \begin{cases} x = a(\theta - \sin \theta) \\ y = a(1 - \cos \theta) \end{cases}$$

$\theta$   $x$   $y$   
0 0 0



A  
B

1. Find the length of the cycloid from one cusp to the next  
 cusp  $x = a(\theta + \sin\theta)$ ,  $y = a(1 - \cos\theta)$ .

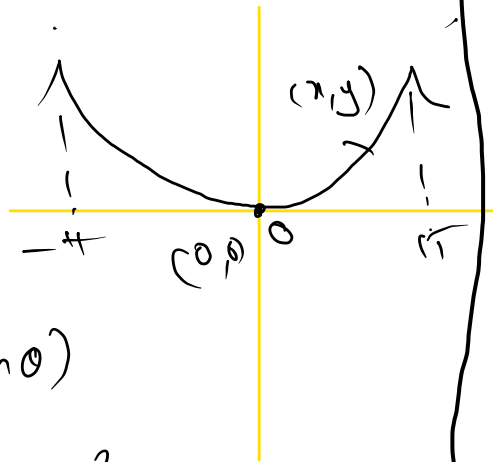
2. If  $S$  is the length of the arc from the origin to a point  
 $p(x, y)$  show that  $s^2 = 8ay$ .

$$S = \int_0^\theta \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \quad \text{--- (1)}$$

$$\frac{dx}{d\theta} = a(1 + \cos\theta) \quad \frac{dy}{d\theta} = a(\sin\theta)$$

$$\begin{aligned} \left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 &= a^2(1 + \cos\theta)^2 + a^2 \sin^2\theta \\ &= a^2[1 + 2\cos\theta + \cos^2\theta + \sin^2\theta] \\ &= a^2[2 + 2\cos\theta] = 2a^2(1 + \cos\theta) \\ &= 2a^2 \cdot 2\cos^2\theta/2 \end{aligned}$$

$$\begin{aligned} S &= \int_{-\pi}^{\pi} \sqrt{4a^2 \cos^2\theta/2} d\theta \\ &= \int_{-\pi}^{\pi} 2a \cos\theta/2 d\theta = 2a \left[ \frac{\sin\theta/2}{1/2} \right]_{-\pi}^{\pi} \\ &= 2a [2\sin\pi/2 - 2\sin(-\pi/2)] \\ &= 2a [\sin\pi/2 + \sin\pi/2] \\ &= 4a [2] = 8a \end{aligned}$$



as  $p(x, y)$  is any pt.  
 on the curve  
 then cycloid is  
 bet<sup>n</sup> angle 0 &  $\theta$

$$\begin{aligned} S &= \int_0^\theta \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} d\theta \\ &= \int_0^\theta 2a \cos\theta/2 d\theta \\ &= 2a \left[ \frac{\sin\theta/2}{1/2} \right]_0^\theta \\ &= 2a [2\sin\theta/2] \\ &= 4a \sin\theta/2 \\ S^2 &= 16 a^2 \sin^2\theta/2 \\ &= 16 a^2 \left( \frac{1 - \cos\theta}{2} \right) \\ &= 8a(a(1 - \cos\theta)) \\ &= 8ay \end{aligned}$$

$r = f(\theta)$  Cardiode

$$r_1 = a(1 + \cos \theta)$$

$$r_2 = a(1 - \cos \theta)$$

$$r = a(1 + \sin \theta)$$

$$r = a(1 - \sin \theta)$$

| $\theta$ | $\cos \theta$ |
|----------|---------------|
| 0        | 1             |
| $\pi/2$  | 0             |
| $\pi$    | -1            |
| $3\pi/2$ | 0             |
| $2\pi$   | 1             |

~~$$r_1 = a(1 + \cos \theta)$$~~

$$r_1 = 2a$$

$$r_2$$

$$0$$

$$a$$

$$a$$

$$2a$$

$$\pi$$

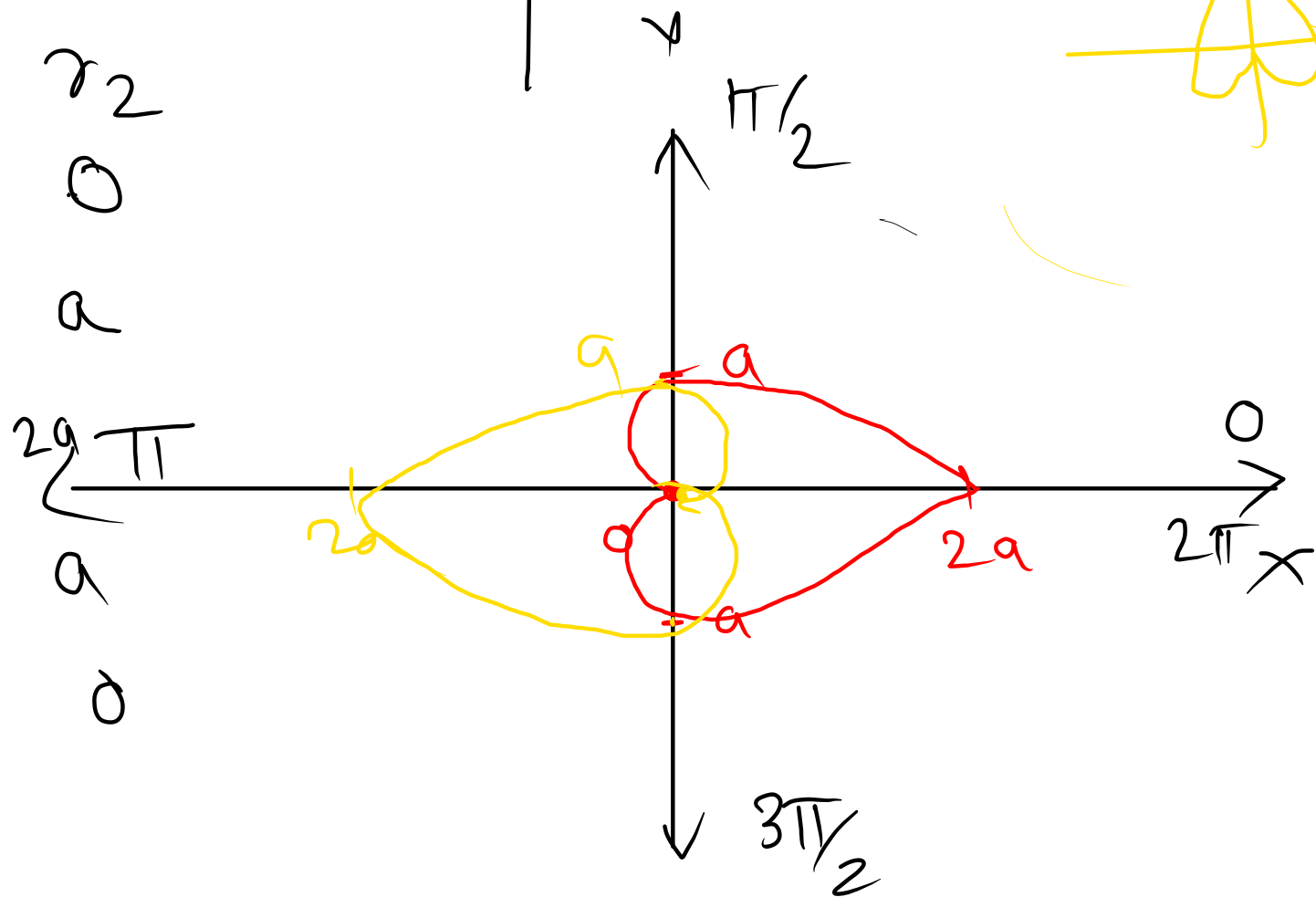
$$a$$

$$0$$

$$a$$

$$2a$$

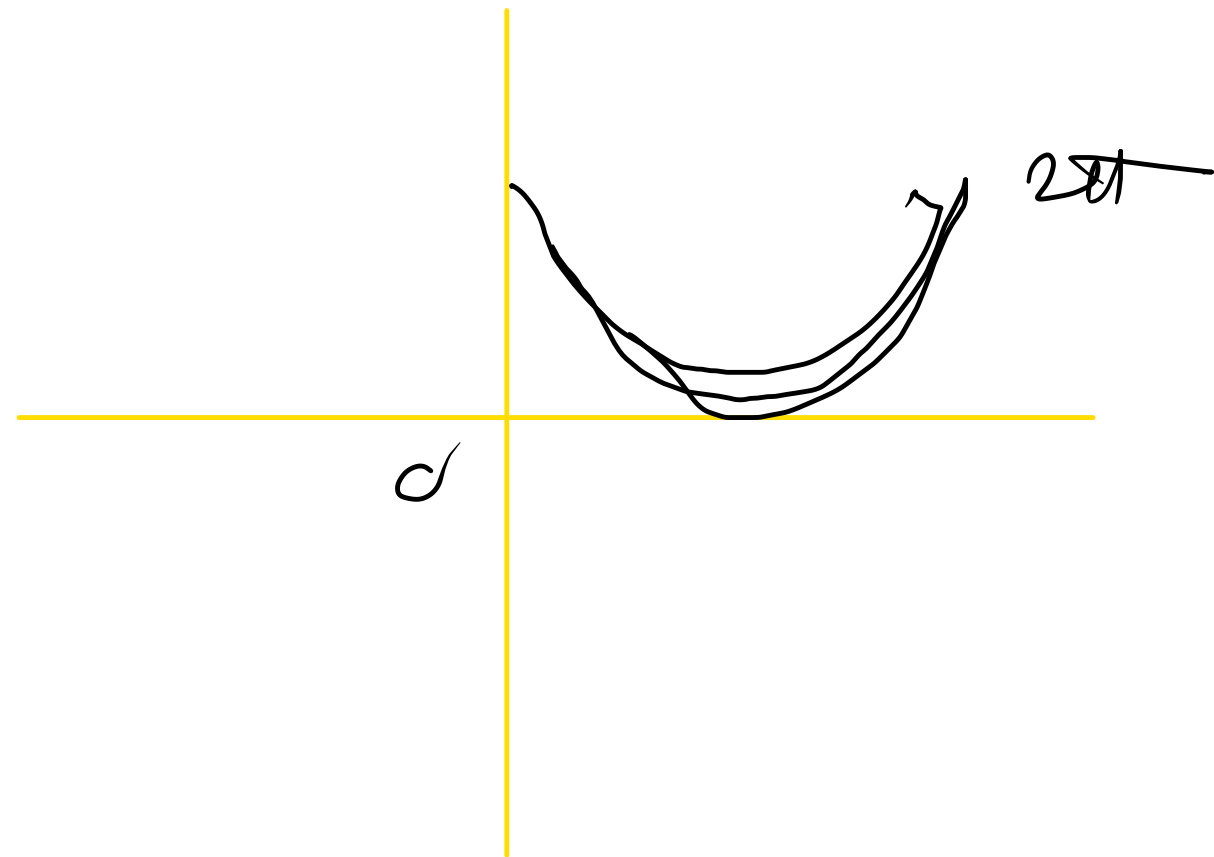
$$0$$





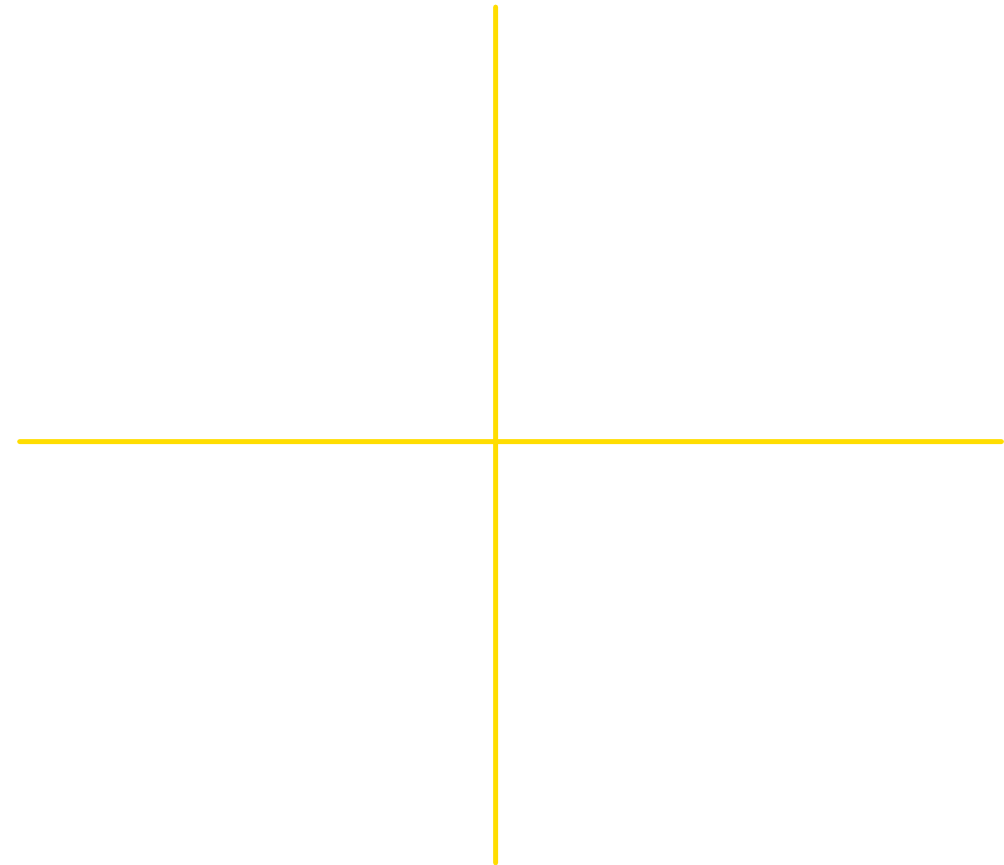
Hw Find the length of the one arc of the Cycloid  $x = a(\theta - \sin \theta)$ ,  $y = a(1 + \cos \theta)$

| $\theta$ | $x$ | $y$ |
|----------|-----|-----|
| 0        | 0   | 2   |



❖ HW Prove that the length of the arc of the curve

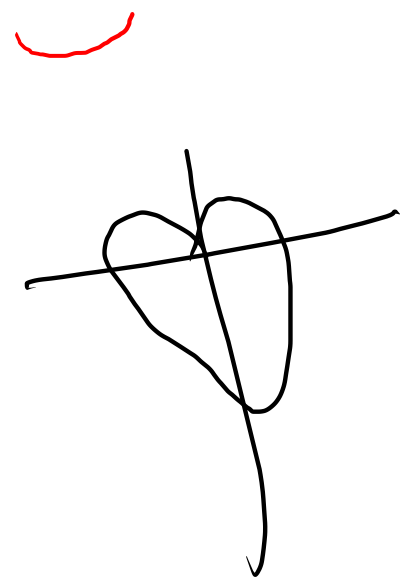
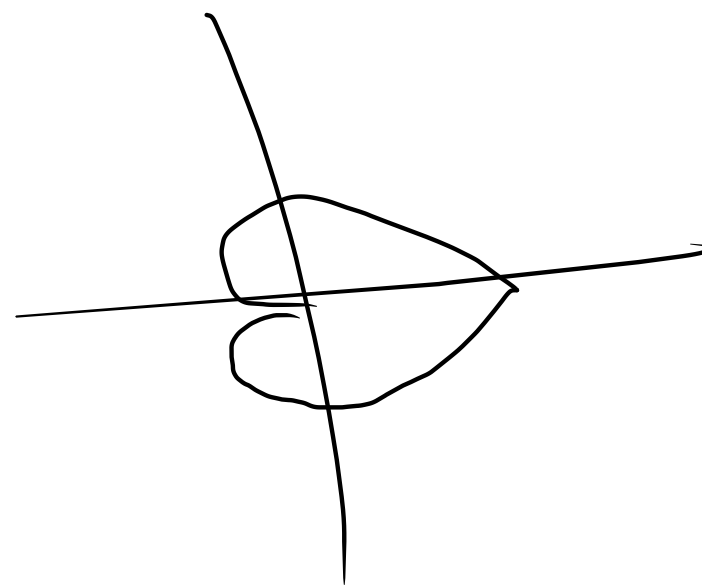
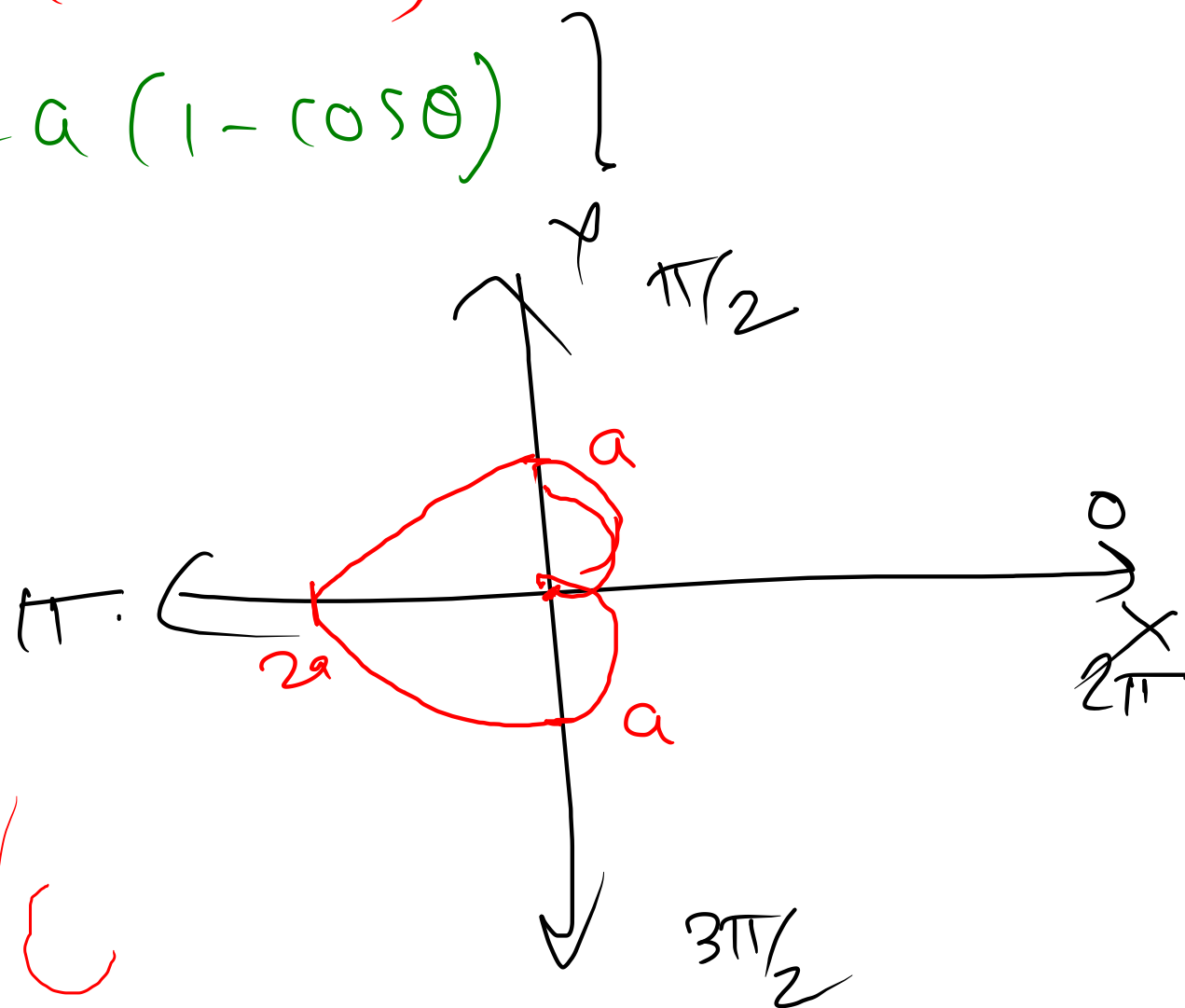
$x = a \sin 2\theta (1 + \cos 2\theta)$ ,  $y = a \cos 2\theta (1 - \cos 2\theta)$  measured from the origin to  $(x, y)$  is  $\frac{4}{3} a \sin 3\theta$ .



$$r = a(1 + \cos \theta)$$

$$\left\{ \begin{array}{l} r = a(1 + \cos \theta) \\ r = a(1 - \cos \theta) \end{array} \right.$$

| $\theta$ | $\cos \theta$ | $r$ |
|----------|---------------|-----|
| 0        | 1             | 0   |
| $\pi/2$  | 0             | a   |
| $\pi$    | -1            | 2a  |
| $3\pi/2$ | 0             | a   |
| $2\pi$   | 1             | 0   |



❖ Find the perimeter of the cardioid  $r = a(1 - \cos \theta)$  and prove that the line  $\theta = \frac{2\pi}{3}$  bisects the upper half of the cardioid.

| $\theta$ | $\cos \theta$ | $r$ |
|----------|---------------|-----|
| 0        | 1             | 0   |
| $\pi/2$  | 0             | a   |
| $\pi$    | -1            | 2a  |
| $3\pi/2$ | 0             | a   |
| $2\pi$   | 1             | 0   |

$S$  = perimeter of cardioid

$= 2$  (upper half of cardioid)

$$= 2 \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$= 2 \int_0^\pi \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

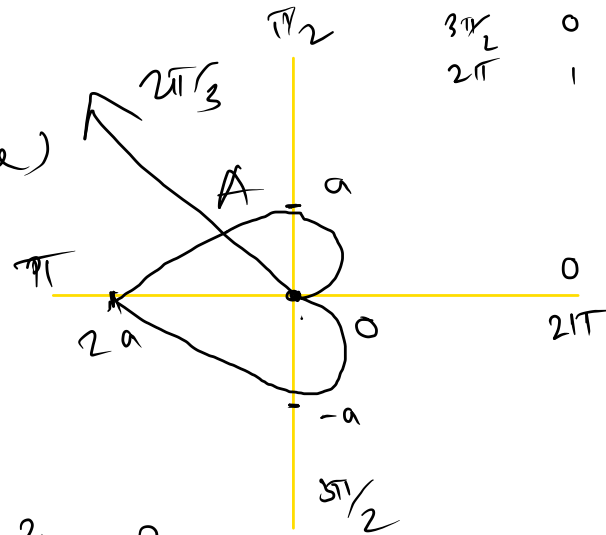
$$\frac{dr}{d\theta} = a(+\sin \theta) \Rightarrow r^2 + \left(\frac{dr}{d\theta}\right)^2 =$$

$$\begin{aligned} &= a^2(1 - \cos \theta)^2 + a^2 \sin^2 \theta \\ &= a^2 [1 - 2\cos \theta + \underbrace{\cos^2 \theta + \sin^2 \theta}_1] \\ &= a^2 [2 - 2\cos \theta] = 2a^2(1 - \cos \theta) \\ &= 4a^2 \sin^2 \theta/2 \end{aligned}$$

$$\begin{aligned} S &= 2 \int_0^\pi 2a \sin \theta/2 = 4a \left[ \frac{\cos \theta/2}{1/2} \right]_0^\pi \\ &= 4a [0 + 2] = 8a // \end{aligned}$$

perimeter of upper half  $= 4a$

$$\begin{aligned} \therefore l(OA) &= \int_0^{2\pi/3} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \\ &= \int_0^{2\pi/3} 2a \sin \theta/2 d\theta = 2a \left[ \frac{-\cos \theta/2}{1/2} \right]_0^{2\pi/3} \\ &= 2a \left[ -\frac{1}{2} \times 2 + 2 \right] \\ &= 2a // \end{aligned}$$



$$(x-a)^2 + (y-b)^2 = r^2$$

❖ Find the length of the cardioid  $r = a(1 - \cos \theta)$  lying inside the circle  $r = a \cos \theta$ .

$$x = r \cos \theta$$

$$y = x \sin \theta$$

$$r = a \cos \theta$$

$$r^2 = a^2 \cos^2 \theta$$

$$x^2 + y^2 = ax$$

$$x^2 - ax + \frac{a^2}{4} + y^2 = \frac{a^2}{4}$$

$$(x - a_2)^2 + y^2 = (a_2)^2$$

Circle: center  $(a_2, 0)$  radi:  $a_2$

$$S = \text{length of cardioid lying inside circle}$$

$$= 2 \times \text{length of cardioid inside circle in 1st Q.}$$

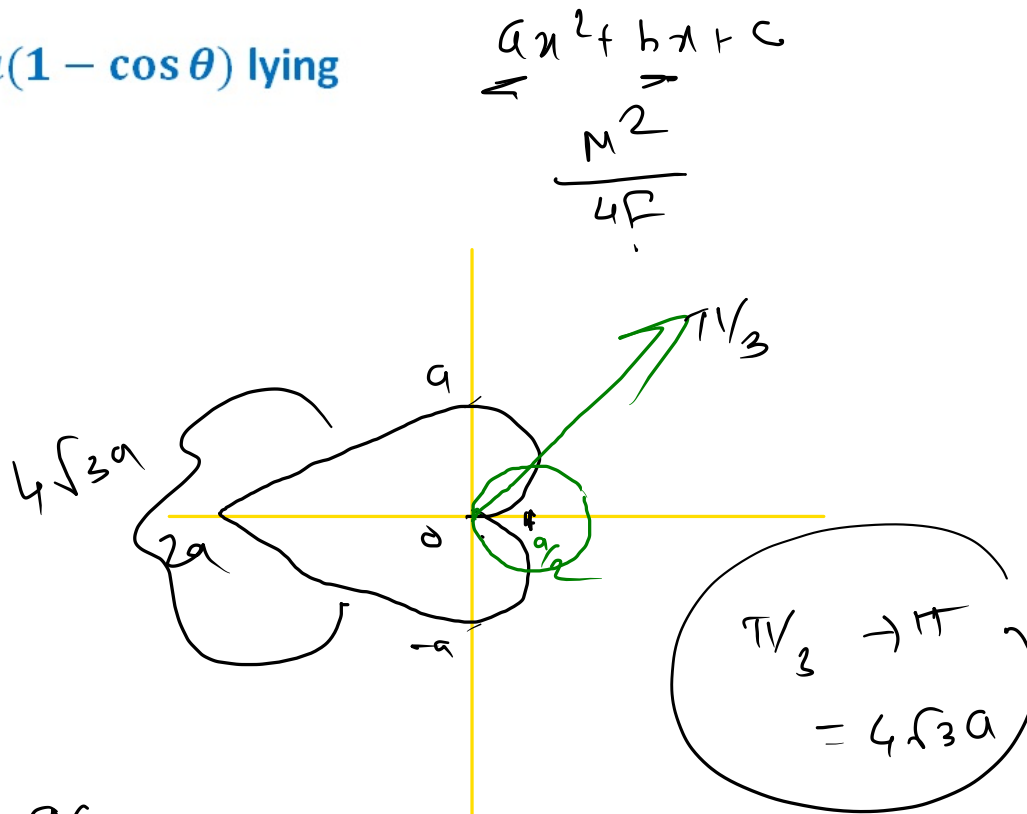
$$= 2 \int_0^{\pi/2} \int_0^{\sqrt{2} \cos \theta} r^2 + \left( \frac{dr}{d\theta} \right)^2 d\theta$$

$$r = a(1 - \cos \theta)$$

$$r = a \cos \theta$$

$$a(1 - \cos \theta) = a \cos \theta \Rightarrow 2 \cos \theta = 1 \Rightarrow \cos \theta = \frac{1}{2}$$

$$S = 2 \int_0^{\pi/3} 2a \sin \theta/2 \, d\theta = 4a \left[ -\frac{\cos \theta/2}{1/2} \right]_0^{\pi/3} = 4a \left[ -\frac{\sqrt{3}}{2} + 2 \right] = 4a(2 - \sqrt{3})$$



lemniscate



❖ Show that the perimeter of  $r^2 = a^2 \cos 2\theta$  is  $\frac{a}{\sqrt{2\pi}} \left( \frac{\pi}{4} \right)^2$

$$r^2 = a^2 \cos 2\theta$$

$$r = \pm a \sqrt{\cos 2\theta} \quad (1)$$

$$\left. \begin{array}{l} \theta = 0 \\ \theta = \pi \end{array} \right\} r = a$$

$$\theta = \pi/4 \quad r = 0$$

$$S = \int_{\theta_1}^{\theta_2} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$= 4 \int_0^{\pi/4} \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta$$

$$= 4 \int_0^{\pi/4} \frac{a}{\sqrt{\cos 2\theta}} d\theta$$

$$= 4a \int_0^{\pi/4} (\cos 2\theta)^{-1/2} d\theta$$

$$2\theta = t \Rightarrow d\theta = dt/2$$

$$= 2a \int_0^{\pi/2} (\cos t)^{-1/2} \frac{dt}{2} \quad \theta: 0 \rightarrow \pi/4 \quad t: 0 \rightarrow \pi/2$$

$$= 2a \cdot \frac{1}{2} B\left(-\frac{1}{2}, \frac{1}{4}\right)$$

$$= a \frac{\Gamma_{1/2} \Gamma_{1/4}}{\Gamma_{3/4}}$$

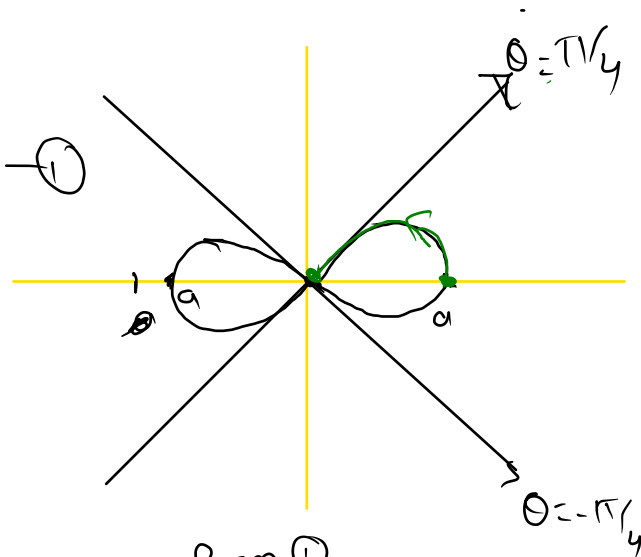
$$= a \sqrt{\pi} \frac{\Gamma_{1/4}}{\Gamma_{3/4}}$$

$$= a \sqrt{\pi} \left(\frac{\Gamma_{1/4}}{\Gamma_{3/4}}\right)^2$$

$$= \frac{a(\sqrt{\Gamma_{1/4}})^2}{\sqrt{2\pi}}$$

$$= \frac{a(\sqrt{\Gamma_{1/4}})^2}{\sqrt{2\pi}}$$

$$= \frac{a(\sqrt{\Gamma_{1/4}})^2}{\sqrt{2\pi}}$$



from (1)

$$\frac{dr}{d\theta} = \frac{a}{2\sqrt{\cos 2\theta}} (-\sin 2\theta) \cdot 2$$

$$r^2 + \left(\frac{dr}{d\theta}\right)^2 = a^2 \cos 2\theta + \frac{a^2 \sin^2 2\theta}{\cos 2\theta}$$

$$= \frac{a^2 (\cos^2 2\theta + \sin^2 2\theta)}{\cos 2\theta}$$

$$= \frac{a^2}{\cos 2\theta}$$

$$\int_0^{\pi/2} \sin^p \theta \cos^q \theta d\theta$$

$$= \frac{1}{2} B\left(\frac{p+1}{2}, \frac{q+1}{2}\right)$$

$$\frac{\Gamma_{1/4} \Gamma_{3/4}}{\Gamma_{1/2}} = \sqrt{2} \pi$$

$$\Gamma_p \Gamma_{1-p} = \frac{\pi}{\sin p\pi}$$

$$\Gamma_{1/4} \Gamma_{3/4} = \frac{\pi}{\sin \pi/4}$$

